

3	(a) electric field strength is force per unit positive charge	B1	[1]
	(b) mass = volume $\times$ density (any subject, allow usual symbols or defined symbols)	C1	
	$= \frac{4}{3} \times \pi \times (1.2 \times 10^{-6})^3 \times 930 (= 6.73 \times 10^{-15})$		
	weight = $\frac{4}{3} \times \pi \times (1.2 \times 10^{-6})^3 \times 930 \times 9.81 = 6.6 \times 10^{-14} \text{ N}$	M1	[2]
(c)	(i) $E = 1.9 \times 10^3 / 14 \times 10^{-3}$	C1	
	$= 1.4 (1.36) \times 10^5 \text{ V m}^{-1}$	A1	[2]
	(ii) $F = QE$		
	$Q = 6.6 \times 10^{-14} / 1.36 \times 10^5$	C1	
	$= 4.9 (4.86) \times 10^{-19} \text{ C}$ [allow $4.7 \times 10^{-19} \text{ C}$ if 1.4×10^5 used]	A1	[2]
	(iii) <u>electric</u> force increases/is greater (than weight)	B1	
	charge (on S) is negative to give resultant/net/sum/total force up	B1	[2]